# Zhang 2021 RCC — analysis summary

RUN 1 OF 4

## GO / NO-GO GATES

### Gate 1 — mixing & continuity → GO

7/8 samples (≥369 cells, dominant 710→192 cliff) intermix on one continuous manifold: iLISI 1.00→3.17, single kNN component (0 detached) at every θ; no patient-private island (kNN-purity 4.6× vs 4.5×, no standoff).

### Gate 2 — module geography → GO (non-branching)

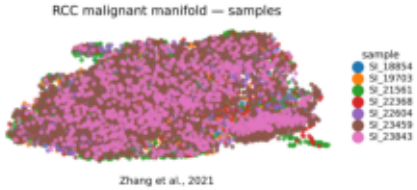
Single developmental axis: pSom-high origin → neural-crest terminus. Crest↔pSom spatially anti-correlated (r=−0.91); Crest rises along pseudotime (ρ=+0.48, high-decile pt 0.72 = terminal); joint pSom&Crest enrichment 0.26 (<1 = inverse gradient, not a co-expression hub). No Gast/pSom arms, no bifurcation. Crest terminal, no inversion flag.

### Gate 3 — hub mutual exclusivity → NA

Not tested — no bifurcation and no hub to run 04_correlation/05_modules on (branching-only gate).

## STEP FIGURES

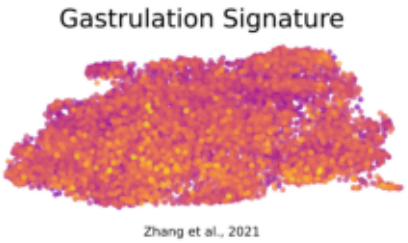
### Manifold — samples



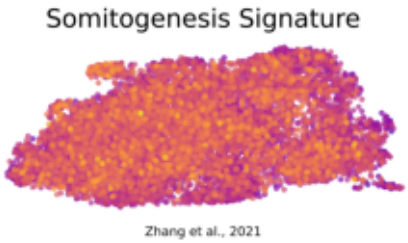

### Pseudotime



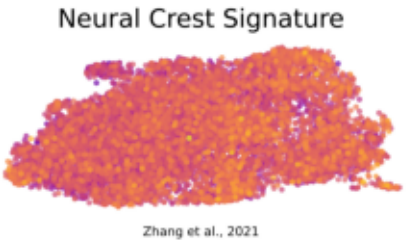

### Gastrulation



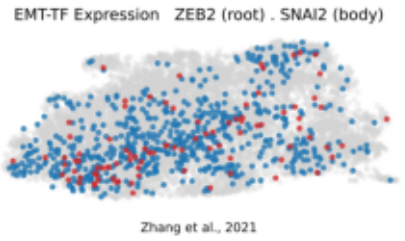

### Somitogenesis




### Neural Crest




### EMT-TF expression




---

**Dataset.**
Clear-cell renal cell carcinoma (Zhang et al., 2021), 10x — 8 patients / 11,410 malignant load; 11,218 across 7 samples (≥369 cells) enter the manifold.

**Verdict.**
Pattern reproduces (non-branching). Gate 1 GO, Gate 2 GO, Gate 3 NA: RCC malignant cells form one continuous developmental axis — posterior-somitogenesis origin → neural-crest terminus, Crest inverse gradient. EMT programme is root-loaded: ZEB2 (24% det.) concentrates at the pSom origin; no terminus EMT-TF.

## Parameter choices

| | | | |
|---|---|---|---|
| **Min cells/sample** | ≥369 (geom-mean cliff; keeps 7/8) | **Island removal** | none (kNN-purity, no standoff) |
| **Root** | tip 0 = pSom-high / Crest-low origin | **Geometry** | non-branching (curve 2 tips 0 forks) |
| **Subtypes** | 3CA meta-programs (mp_top) | **Integration** | Harmony on sample, θ=10, 30 it |
| **Trajectory** | scFates curve N=20, λ=0.3, μ=100 | **n_pcs** | 20 (10x precedent; scree top-heavy) |
| **Signatures** | score_genes, union control pool | **EMT-TFs** | ZEB2 (root) / SNAI2, root-loaded |